

Wireless Access Electromagnetics

Assignment 02

Sheharyar Ahmed Nizamani

S329580

Exercise 1

Solution

Sheharyar Ahmed Nizamani
5329580

11

Assignment #02
WAE

Pb#01

Given, $Z_{r1} = Z_{r2} = Z_{r3} = Z_0 = 50\Omega$

$\epsilon_r = 4.45$

$S_{11} = 0$

$h = 1.55 \text{ mm}$

$f_0 = 5.8 \text{ GHz}$

$\frac{P_3}{P_2} = \alpha$; $\alpha = 1.25$

Sol

$$\frac{P_3}{P_2} = \frac{Z_{\infty 3}}{Z_{\infty 2}} = \alpha \rightarrow Z_{\infty 3} = \alpha Z_{\infty 2}$$

$$\frac{Z_0}{Z_{\infty 2}} + \frac{Z_0}{Z_{\infty 3}} = \frac{1}{Z_0} \rightarrow \frac{1}{Z_{\infty 2}} + \frac{1}{\alpha Z_{\infty 2}} = \frac{1}{Z_0}$$

$$\alpha Z_{\infty 2} = (1 + \alpha) Z_0^2$$

$$Z_{\infty 2} = \frac{1 + \alpha}{\alpha} Z_0^2 = \frac{1 + 1.25}{1.25} (50)^2 = 4500$$

$$Z_{\infty 2} = 67.08\Omega$$

$$Z_{\infty 3} = \alpha Z_{\infty 2} \Rightarrow 1.25 \times 4500 = 5625$$

$$\boxed{Z_{\infty 3} = 75\Omega}$$

$$Z_{\infty 2} \rightarrow \frac{W_2}{h} = 1.114 \rightarrow W_2 = 1.73 \text{ mm}$$

$$A = \frac{67.08}{60} \sqrt{\frac{4.45+1}{2}} + \frac{4.45-1}{4.45+1} \left(0.23 + \frac{0.11}{4.45} \right) = 2.007$$

$$\frac{W}{h} < 2 = \frac{8e^A}{e^{2A}-2} = \frac{8e^{2.007}}{e^{2 \times 2.007} - 2} = 1.114$$

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 2 \frac{h}{W}}} \Rightarrow \frac{4.45+1}{2} + \frac{4.45-1}{2} \frac{1}{\sqrt{1 + 2 \left(\frac{1.55}{1.73} \right)}} = 3.22$$

$$\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8}{5.8 \times 10^9} = 0.0517 \quad (2)$$

$$l_2 = \frac{\lambda_0}{\sqrt{\epsilon_{eff_2}}} = \frac{0.0517}{\sqrt{3.22}} = 2.88 \text{ cm}$$

$$l_2 = l_2 / 4 = 0.72 \text{ cm}$$

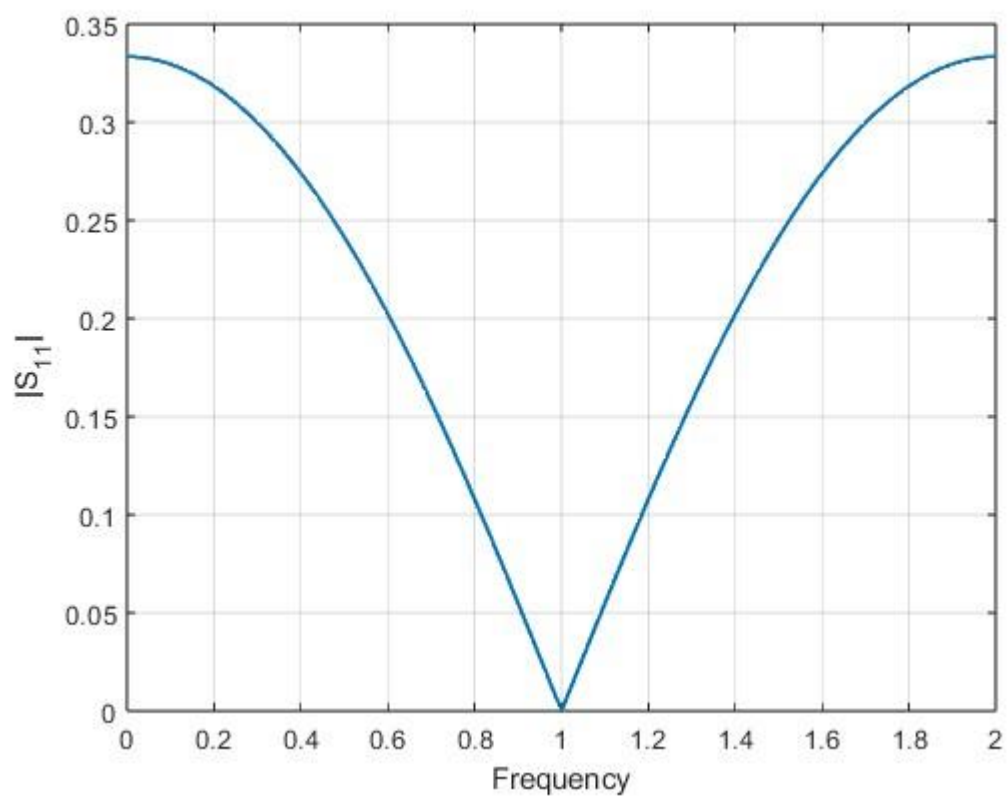
$$Z_{\infty_3} \rightarrow \frac{W_3}{h} = 0.89 \rightarrow W_3 = 1.37$$

$$\epsilon_{eff_3} = 3.18 \rightarrow l_3 = \frac{0.0517}{\sqrt{3.18}} \Rightarrow 2.9 \rightarrow l_3 = 0.724$$

$$Z_0 = \frac{W_0}{h} = \frac{1.90}{1.893} \rightarrow W_0 = 2.935$$

- Comments

To design the T-junction, first I determined how to split the input admittance between the two branches to meet the specific power ratio. After calculating the required impedances, I translated those values into microstrip widths and lengths suited for the dielectric material. Finally, I ran a MATLAB simulation to confirm that the reflection coefficient hits zero exactly at the design frequency.



Exercise 2

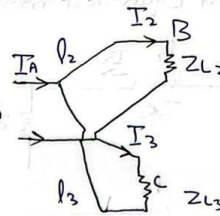
Solution

Pb #02:

Sol: For $S_{11} = 0$, $\frac{1}{Z_{02}} + \frac{1}{Z_{03}} = \frac{1}{Z_{01}}$ A

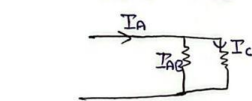
If $Z_{02} = Z_{L2}$ and $Z_{03} = Z_{L3}$,

$$\Rightarrow \frac{1}{112.5} + \frac{1}{90} = \frac{1}{50}$$

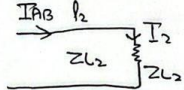


$$|I_3| = 1.25 |I_2|, |I_2| = \frac{|V_j|}{Z_{02}}, |I_3| = \frac{|V_j|}{Z_{03}}$$

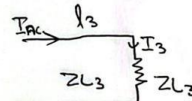
$$\frac{|I_3|}{|I_2|} = \frac{Z_{02}}{Z_{03}} = \frac{112.5}{90} = 1.25$$



$$I_A = I_{AB} + I_{AC}$$



$$I_2 = I_{AB} e^{jk_2 l_2}$$



$$I_3 = I_{AC} e^{jk_3 l_3}$$

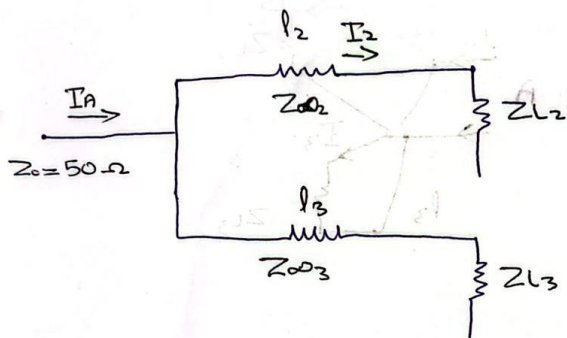
$$\angle I_3 - \angle I_2 = -\pi \Rightarrow -k_3 l_3 + k_2 l_2 = -\pi$$

$$k_3 = k_2 \Rightarrow -\frac{2\pi}{\lambda_0} (l_3 - l_2) = -\pi \Rightarrow l_3 - l_2 = \frac{\lambda_0}{2}$$

$$\text{if } \angle I_2 = -\frac{\pi}{2} \Rightarrow k_2 l_2 = \frac{\pi}{2} \Rightarrow l_2 = \frac{\lambda_0}{4}$$

$$\text{Substituting, } l_3 - \frac{\lambda_0}{4} = \frac{\lambda_0}{2} \Rightarrow l_3 = \frac{3\lambda_0}{4}$$

T-Junction Equivalent model:

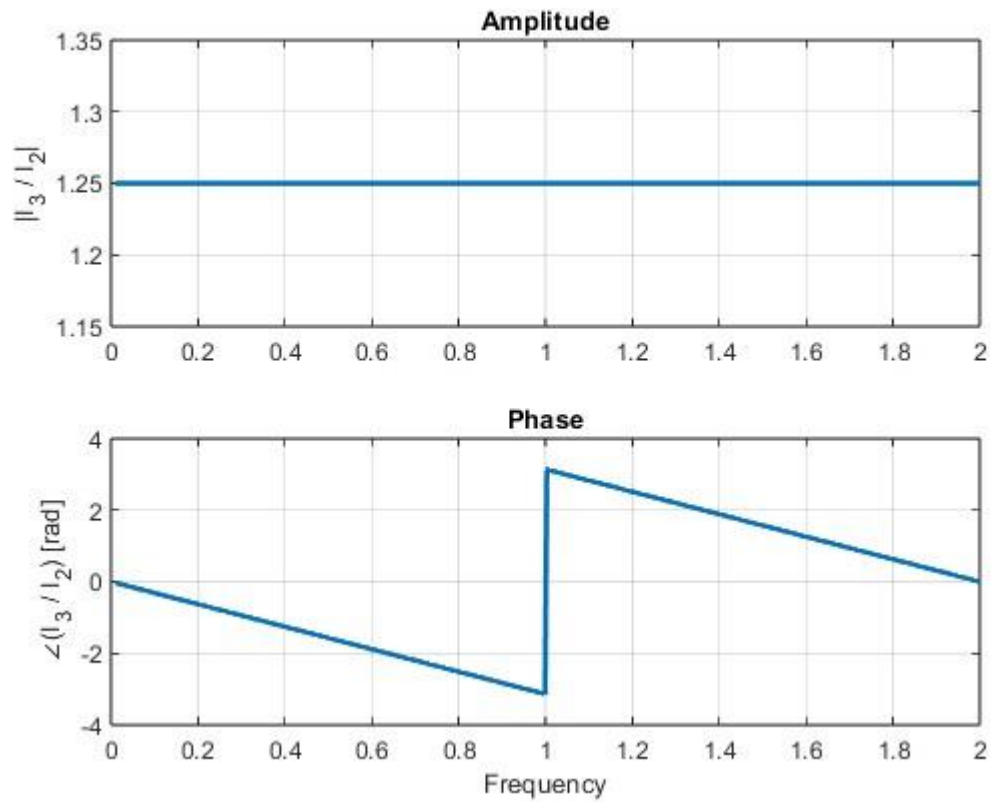


CS Scanned with CamScanner

- Comments:

I chose to match the characteristic impedance of each branch directly to its specific load. This approach was effective because it keeps the input impedances constant, which makes it easier to satisfy the matching condition at the junction. I adjusted the physical lengths of the two lines so that their electrical difference results in exactly a 180-degree shift at 5.8 GHz. The MATLAB results

confirm this logic, showing a steady current ratio of 1.25 across the band and the correct phase reversal at the center frequency.



MATLAB Script

Exercise 1:

```
clear all;
close all;
clc
f0 = 5.8e9;
f = linspace(0.1, 2*f0, 500);
Zr = 50;
Zc2 = 75;
Zc3 = sqrt(4500);
theta = (pi/2) * (f/f0);
Zin2 = Zc2 * (Zr + 1j*Zc2*tan(theta)) ./ (Zc2 + 1j*Zr*tan(theta));
Zin3 = Zc3 * (Zr + 1j*Zc3*tan(theta)) ./ (Zc3 + 1j*Zr*tan(theta));
Zin_total = (Zin2 .* Zin3) ./ (Zin2 + Zin3);
S11 = (Zin_total - Zr) ./ (Zin_total + Zr);
figure;
plot(f/f0, abs(S11), 'LineWidth', 1.5);
grid on; xlabel('Frequency'); ylabel('|S_{11}|');
```


Exercise 2:

```
clear all;
close all;
clc
f0 = 5.8e9;
Zc2 = 112.5; ZL2 = 112.5;
Zc3 = 90; ZL3 = 90;
fn = linspace(0.01, 2, 500);
ratio = zeros(size(fn));
for i = 1:length(fn)
    theta2 = (pi/2) * fn(i);
    theta3 = (3*pi/2) * fn(i);
    den2 = ZL2 * cos(theta2) + 1j * Zc2 * sin(theta2);
    den3 = ZL3 * cos(theta3) + 1j * Zc3 * sin(theta3);
    ratio(i) = den2 / den3;
end
figure;
subplot(2,1,1);
plot(fn, abs(ratio), 'LineWidth', 2);
grid on;
ylim([1.15 1.35]);
ylabel('|I3 / I2|');
title('Amplitude');
subplot(2,1,2);
plot(fn, angle(ratio), 'LineWidth', 2); grid on;
ylabel('\angle(I3 / I2) [rad]'); xlabel('Frequency');
title('Phase');
```